

CURRICULUM VITAE

Niels Pacheco-Barrios MD, Msc

PERSONAL DATA

Home Address (Peru): Av. Aurelio García y García 1636 – Cercado de Lima, Lima, Peru

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Date of Birth: 10/17/1997

DEGREES

Institution and location	Degree	Completion Date	Field of Study
Universidad Peruana Cayetano Heredia, Peru	MBBS	05/2022	Medicine
Universidad Peruana Cayetano Heredia, Peru	MD	05/2022	Medicine
Dresden International University, Germany	M.Sc.	02/2024	Biostatistics and Bioinformatics
Harvard Medical School, USA	M.M.Sc	05/2025	Clinical Investigation
Harvard Medical School, USA Global Clinical Scholars Research Training (GCSRT)	PG. Dip	06/2023	Epidemiology and Biostatistics
Harvard T. Chan School of Public Health Principles and Practice of Clinical Research (PPCR)	PG. Dip	11/2022	Epidemiology and Biostatistics

PROFESIONAL EXPERIENCE AND OTHER EDUCATION

09/2025 - Current	The Mapping & Engineering Neural Dynamics (MEND) Lab, Harvard Medical School – Boston, USA. Post-doctoral Research Fellow.
08/2023 – 05/2025	The Mapping & Engineering Neural Dynamics (MEND) Lab, Harvard Medical School – Boston, USA. Graduate (Master's) Student
11/2024 – Current	Pehlevan Group Theory of neural networks for natural and artificial intelligence – Harvard School of Engineering and applied science (SEAS) – Boston, USA. Research Collaborator
08/2025 - Current	MIT InfoLab Language, Perception and Robotics – MIT Computer Science and Artificial Intelligence Laboratory (CSAIL) – Boston, USA, Research Collaborator
05/2024 – Current	Visiting Professor in Epidemiology and Biostatistics – Universidad Científica del Sur, Lima, Peru
05/2024 – Current	Lecturer in Neuropharmacology and Biostatistics – Universidad Peruana Cayetano Heredia, Lima, Peru
02/2023 – 09/2023	Health technology assessment consultant – IETSI EsSalud – Lima, Peru
01/2019 - 02/2019	Spaulding Neuromodulation Center, Harvard Medical School – Boston, USA. Research Observership
02/2016 – 05/2022	CRONICAS Centre of Excellence in Chronic Diseases – Lima, Peru. Research trainee.
02/2016 – 05/2022	EMERGE Centro de investigación en enfermedades emergentes y cambio climático – Lima Peru. Research Trainee
02/2016 – 05/2022	REDECS Red de eficacia clínica – Lima, Perú. Research Trainee
01/2021 – 02/2021	Doctor Hospital at Renaissance – Texas, USA. General Surgery Observership
02/2021 – 03/2021	Doctor Hospital at Renaissance – Texas, USA. Neurology Observership
03/2021 – 05/2021	Doctor Hospital at Renaissance – Texas, USA. Neurosurgery Observership
01/202 – 02/2020	Doctor Hospital at Renaissance – Texas, USA. Cardiology Observership
07/2019	University of Sao Paulo, Winter School 2019 – Sao Paulo, Brazil. Neurology Clerkship
02/2025 – 05/2025	HST 582/6.8800 Biomedical Signal and Image processing – Harvard/ MIT – Boston, USA
09/2024 – 12/2024	APMTH 226 Theory of Neural Computation – Harvard SEAS – Boston, USA
09/2024 – 12/2024	HST 953/6.8850 Clinical Data Machine Learning – Harvard/ MIT – Boston, USA
07/2024	Computational Neuroscience course - Neuromatch
06/2024	Advance methods for Confounding adjusment – Harvard T. Chan School of Public Health, Causal Lab – Boston, USA
06/2024	Target Trial Emulation – Harvard T. Chan School of Public Health, Causal Lab – Boston, USA
12/2024	MATLAB Signal Processing - MATLAB self-paced courses
07/2023	MATLAB Onrapm - MATLAB self-paced courses
12/2023	MATLAB Fundamentals - MATLAB self-paced courses
03/2023 – 07/2023	Python Specialization – DMC Institue – Lima, Peru
01/2023 – 03/2023	Python Fundamentals – DMC Institue – Lima, Peru
02/2023 – 03/2023	SQL server Fundamentals – DMC Institue – Lima, Peru
12/2022 – 03/2023	Data Science with R – Data & Analytics Academy – Lima, Peru
12/2022 – 03/2023	Data Science with Python – Data & Analytics Academy – Lima, Peru
01/2023 – 02/2023	Biostatistics with Rstudio – Evisalud – Lima, Peru
07/2020 – 08/2029	Biostatistics with STATA – Evisalud – Lima, Peru
07/2022 – 07/2023	IBM Python Specialization - Coursera
03/2018 – 07/2018	Scientific writing, Epidemiology and Biostatistics – IETSI, EsSalud – Lima, Peru

03/2019 – 06/2019	STATA for Biostatistics – Universidad Peruana Cayetano Heredia – Lima, Peru
07/2018 – 12/2018	Biostatistics – Universidad Peruana Cayetano Heredia – Lima, Peru

HONORS AND AWARDS

- Winner of the **National Science Foundation Scholarship** to attend the **Brain, Minds and Machines** Summer Course at the Marine Biological Laboratory from the University of Chicago in August 2025.
- **Winner of the Kaelin grant at Essalud 2024** – A research grant to develop a chatbot to aid in diagnosing and treating mental health disorders among patients with tuberculosis undergoing telehealth follow-up.
- **Winner of the scholarship “Presidente del Peru – Bicentenario” 2023**, an educational grant for international postgraduate education. Covers 80K+ for a master’s degree in clinical investigation at Harvard Medical School.
- **Winner of the Harvard Medical School Dean Scholarship 2023** – a 50% discount for postgraduate studies in Clinical Research
- **Winner of the Scholarship for *Multidisciplinary Training Program on Neuropsychiatry and Behavioral Disorders in First Nations (NEUFIN) 2024*** – Funding to be part of the fellowship in Neuroscience, where we receive training in neuroimaging, neuroanatomy, neurophysiology and systems neuroscience to be applied in research in Quechua populations.
- **Winner of the Social responsibility grant by Cayetano Heredia Peruvian University (2018)**. For the creation of TB out, an educational program to educate about and prevent tuberculosis.
- **Winner of the Tejada Scholarship (2021)** for the project “Association of the hydric and caloric balance with mortality in patients with COVID-19 patients at the ICU department in a major hospital in Lima, Peru”.
- **Winner of the Student research grant, Cayetano Heredia Peruvian University (2017)** for the project “Association of work performance and HIV-neurocognitive disorder in a public hospital in Lima, Peru”.
- **Innovate4health 2020 - 2021 by John Hopkins University**. Selected as the winner team among 50 teams worldwide, trained by the innovation incubator for Emergent infectious diseases from John Hopkins. We developed a web-based system to integrate and monitor infectious diseases in peripheral and rural areas of Peru, creating an information system to deliver trust in healthcare among the population.
- **Honorific mention at the Global Hackathon Mission Brain 2023 (March 24-26, 2023)**. Munay 4 Rehab is a web application to provides language therapy to non-affluent aphasia. Team: Niels Pacheco-Barrios, Carlos Vasquez, Alyssa Maguiña.
- **First Place in Original Articles competition** Benites-Gamboa D, Chacón JI, Pacheco-Barrios N, Castillo G, Paucar LS, Villabona S, Mejia CR. Trastornos de control de impulsos y adicciones según universidad de estudiantes de medicina: Multicéntrico de diecinueve sedes en Latinoamérica. **Congreso Científico Internacional (CCI) Bolivia-2016**.
- **First place in Original Articles competition** Tellez WA, Nieto-Gutierrez W, Pacheco-Barrios N, Aguirre-Tipismana L, Zevallos-Morales A, Taype-Rondan A. Características de patentes de dispositivos para rehabilitación física registradas en los Estados Unidos. **Congreso Científico Nacional (CCN) Ica, Perú-2017**.

PUBLICATIONS AND RESEARCH

Google Scholar profile: [Niels Pacheco-Barrios MD Msc - Google Scholar](#)

H-Index: 9

Number of citations: 434

Journal Articles (signifies peer review)

1. Howard CW, Reich M, Luo L, **Pacheco-Barrios N**, Alterman R, Rios AS, et al. Cognitive outcomes of deep brain stimulation depend on age and hippocampal connectivity in Parkinson's and Alzheimer's disease. *Alzheimers Dement* [Internet]. 2025 Aug;21(8):e70498. Available from: <http://dx.doi.org/10.1002/alz.70498>
2. Bai, Y., Pacheco-Barrios, K., **Pacheco-Barrios, N.**, Liang, G., & Fregni, F. (2024). Neurocircuitry basis of motor cortex-related analgesia as an emerging approach for chronic pain management. *Nature Mental Health*, 2(5), 496–513.
3. Schaper, F. L. W. V. J., Morton-Dutton, M., **Pacheco-Barrios, N.**, Turner, J. I., Drew, W., Khosravani, S., Joutsa, J., & Fox, M. D. (2024). Brain lesions causing parkinsonism versus seizures map to opposite brain networks. *Brain Communications*, 6(3), fae196.
4. Hart LA, Warren AE, **Pacheco-Barrios N**, Bahners BH, Madan S, Neudorfer C, et al. Deep brain stimulation for epilepsy: Optimal targeting and clinical outcomes [Internet]. medRxiv. 2025. p. 2025.06.23.25330152. Available from: <https://www.medrxiv.org/content/10.1101/2025.06.23.25330152.abstract>
5. **Pacheco-Barrios N.**, Wadhwa A, Tripathy S, Jha R, Wadhwa M, Warren AEL, et al. The effects of deep brain stimulation on sleep: A systematic review and meta-analysis. *Sleep Adv* [Internet]. 2024 Oct 22 [cited 2024 Nov 13];5(1): zpae079. Available from: <https://academic.oup.com/sleepadvances/article-pdf/5/1/zpae079/60534139/zpae079.pdf>
6. **Pacheco-Barrios N**, Gagliardi V, Grudtner RR, Njokanma I, Illigens B, Rolston JD, et al. Association of chronic pain with motor symptom severity in Parkinson's disease: An exploratory cross-sectional analysis. *Life* [Internet]. 2025 Feb 1;15(2):268. Available from: <http://dx.doi.org/10.3390/life15020268>
7. **Pacheco-Barrios N**, Wadhwa A, Lau TS, Shutran M, Ogilvy CS. Risk Factors Associated with Seizure After Treatment of Chronic Subdural Hemorrhage: A Systematic Review and Meta-Analysis. 2024 Sep 30;10.1227.
8. **Pacheco-Barrios, N.**, Miranda-Chavez, B., & Taype-Rondan, A. (2024). Public interest in palliative care in Latin America: A Google Trends analysis. *PloS One*, 19(7), e0306312.
9. **Pacheco-Barrios N**, Wadhwa A, Purohit S, Enriquez-Marulanda A, Maroufi SF, Alwakaa O, et al. Demographic and social determinants of mortality in cerebral arteriovenous malformations: identifying high-risk clusters in the U.S. *Neurosurg Rev* [Internet]. 2025 Nov 20;49(1). Available from: <http://dx.doi.org/10.1007/s10143-025-03960-w>
10. Bhalla S, Romero B, Al-Nuaimy Y, Toussaint F, Zoghi S, **Pacheco-Barrios N**, et al. Spatial-in-spatial: The utility of combining fluorescence-guided multiple sampling with spatial-omics in human glioblastoma. *Mol Cancer Res* [Internet]. 2025 Jul 25;OF1–5. Available from: <http://dx.doi.org/10.1158/1541-7786.mcr-25-0194/764249>
11. João RB, **Pacheco-Barrios N**, Leite M, Gonçalves Soares YRR, Eylül Bakir Z, Veiga ME, et al. Modafinil/armodafinil for excessive daytime sleepiness after traumatic brain injury: a systematic review and meta-analysis. *Brain Inj* [Internet]. 2025 Aug;39(11):933–41. Available from: <http://dx.doi.org/10.1080/02699052.2025.2502424>
12. Terry F, Enríquez-Marulanda A, Chinchihualpa-Paredes N, Carbajal-Galarza M, Vidal-Cuéllar CL, Mas-Ubillus G, **Pacheco-Barrios N**, et al. Clinical predictors of inpatient mortality and poor postoperative course after aSAH

- microsurgical clipping: A 10-year experience from a Peruvian tertiary care center. *J Clin Med* [Internet]. 2025 Jul 1;14(13):4799. Available from: <http://dx.doi.org/10.3390/jcm14134799>
13. Venero C Jr, Roy JM, Ghurye N, Warriar A, Khalid MU, **Pacheco-Barrios N**, et al. Frailty indices in patients undergoing functional neurosurgical procedures: A systematic review. *Stereotact Funct Neurosurg* [Internet]. 2025 Jul 2;1–10. Available from: <http://dx.doi.org/10.1159/000547128/4398598/000547128>
 14. Andreão FF, Feitosa Filho HN, Barros EA, Abreu Moreira JG, Hemais M, Neto AR, **Pacheco-Barrios N**, et al. Endovascular or microsurgical? Defining the best approach for blood blister aneurysms: A comparative meta-analysis. *Neuroradiol J* [Internet]. 2025 Jun 12;19714009251346467. Available from: <http://dx.doi.org/10.1177/19714009251346467>
 15. Navarro-Flores, Alba*, Jose Ernesto Fernandez-Chinguel*, **Niels Pacheco-Barrios**, David R. Soriano-Moreno, and Kevin Pacheco-Barrios. 2022. “Global Morbidity and Mortality of Central Nervous System Tuberculosis: A Systematic Review and Meta-Analysis.” *Journal of Neurology*, March. **10.1007/s00415-022-11052-8**
 16. Gomes FC, Larcipretti ALL, Udoma-Udofa OC, de Abreu Rocha BA, Mota MEB, Decina MM, **Pacheco-Barrios, N.** et al. Stereoelectroencephalography versus subdural electrodes for invasive monitoring of drug-resistant epilepsy patients: a systematic review and meta-analysis. 2025 Apr 2; Available from: <https://www.sciencedirect.com/science/article/pii/S1059131125000858>
 17. Armas-Salazar A, Alfonso-Gutiérrez LM, Jader A, Avecillas-Chasin JM, Aronson JP, **Pacheco-Barrios N**, et al. Ventral Capsule/Ventral Striatum Deep Brain Stimulation versus Anterior Capsulotomy in Treatment-Resistant Major Depression: A Systematic Review and Data Analysis. 2024 Dec 22;108703. Available from: <https://www.sciencedirect.com/science/article/pii/S0303846724005900>
 18. Ramirez-Velandia, F., Ranawaka, K. H., Wadhwa, A., Salih, M., Fodor, T. B., Lau, T. S., **Pacheco-Barrios, N.**, Enriquez-Marulanda, A., Khan, I. S., Vega, R. A., Stippler, M., Taussky, P., Hong, J., & Ogilvy, C. S. (2024). Comparison of Postoperative Seizures Between Burr-Hole Evacuation and Craniotomy in Patients With Nonacute Subdural Hematomas: A Bi-Institutional Propensity Score-Matched Analysis. *Neurosurgery*. <https://doi.org/10.1227/neu.0000000000003046>
 19. Barhouse, P., Young, M., Taussky, P., **Pacheco-Barrios, N.**, & Ogilvy, C. S. (2024). Anterior choroidal artery aneurysms: a systematic review and meta-analysis of outcomes and ischemic complications following surgical and endovascular treatment. *Journal of Neurosurgery*, 1–11.
 20. Terry F, Orrego-González E, Enriquez-Marulanda A, **Pacheco-Barrios N**, Merenzon MA, Komotar RJ, et al. Temporal dynamics and clinical predictors of brain metastasis in breast cancer: A two-decade cohort analysis toward tailored CNS screening. *Cancers (Basel)* [Internet]. 2025 Mar 1;17(6):946. Available from: <http://dx.doi.org/10.3390/cancers17060946>
 21. Ramirez-Velandia F, Ranawaka KH, Wadhwa A, Fodor TB, Salih M, Mensah EO, **Pacheco-Barrios, N.** et al. Risk factors of postprocedure seizures following standalone middle meningeal artery embolization of chronic subdural hematomas: a bi-institutional retrospective analysis. *Neurosurg Rev*. 2025 Apr 3;48(1):350.
 22. Ranawaka KH, Ramirez-Velandia F, Fodor TB, Wadhwa A, Salih M, Lau TS, **Pacheco-Barrios, N.** et al. Understanding risk factors for postoperative seizure following surgical treatment and middle meningeal artery embolization of chronic subdural hematomas. *World Neurosurg*. 2025 Mar 15;197:123796.
 23. Ramirez-Velandia, F., Wadhwa, A., Mensah, E., Sathya, A., **Pacheco-Barrios, N.**, Filo, J., Pettersson, S. D., Enriquez-Marulanda, A., Young, M., Granstein, J. H., Taussky, P., & Ogilvy, C. S. (2024). Endovascular Treatment of Basilar Apex Aneurysms: An Updated Systematic Review and Meta-Analysis in the Era of Flow Diversion. *World Neurosurgery*. <https://doi.org/10.1016/j.wneu.2024.07.142>
 24. Jagtiani P, Dos Santos K, Iqbal J, Mahto N, Ochieng JAO, **Pacheco-Barrios N**, et al. Fragile fortunes: Optimizing outcomes by identifying high-risk neurosurgical patients through frailty-based risk assessment. 2025 Jan 1;443–64.

25. Barbagli G, Aglan O, Hussein A, Soto-Rubio D, **Pacheco-Barrios N**, Li C, et al. Trigeminal neuralgia as a presentation of high cervical stenosis: a systematic review of the literature. *Eur Spine J* [Internet]. 2025 Feb;34(2):707–18. Available from: <http://dx.doi.org/10.1007/s00586-024-08614-x>
26. Hussein A, Quiceno E, **Pacheco-Barrios N**, Dholaria N, Pico A, Barbagli G, et al. Lumbar disc herniation and cauda equina syndrome during pregnancy: A systematic review. *Acta Neurochir (Wien)*. 2024 Nov 26;166(1):479.
27. Alexandros Moniakakis MD, MSc, **Pacheco-Barrios, N MD, Msc**, Quiceno, E., Hussein, A., Ms, A. P., Abdulla, E., Bauer MS, I. L., Nosova, K., Khan, M. A., Farhadi, D. S., Prim, M., & Baaj, A. (2024). Does Routine Post-Operative Use of Drainage in Minimally Invasive Lumbar Spine Surgery Offer Better Results? *Journal of Spine Practice (JSP)*, 52–60.
28. Gil-Olivares, F., Alva-Díaz, C., Pinedo-Torres, I., **Pacheco-Barrios, N.**, Llerena, K. M. A., & Huerta-Rosario, M. (2024). Postulates of evidence-based medicine have transformed into myths. *Educación Médica*, 25(3), 100887.
29. Marsool Marsool, M. D., Bharadwaj, H. R., Ali, S. H., Aderinto, N., Shah, M. H., Shing, N., Dalal, P., Huang, H., Wellington, J., Chaudri, T., Awuah, W. A., **Pacheco-Barrios, N.**, Macha-Quillama, L., Fernandez-Guzman, D., & Hussien Mohamed Ahmed, K. A. (2024). Exploring the Landscape of Intracranial Aneurysms in South America: A Comprehensive Narrative Review Intracranial Aneurysms in South America. *World Neurosurgery*, 185, 3–25.
30. Gonçalves OR, Dominici S, dos Santos Monteiro J, Weba ETP, Hong A, Santos AB, **Pacheco-Barrios, N.** et al. Stratifying clinically relevant risk factors for acute ischemic stroke and cerebrovascular events in chagas disease: a systematic review and meta-Analysis. 2024 Dec 1;102950. Available from: <https://www.sciencedirect.com/science/article/pii/S0146280624005851>
31. Dholaria, Nikhil, Isabel Bauer, James Kelbert, Giovanni Barbagli, Annemarie Pico, Courtney Deaver, Esteban Quiceno, **Niels Pacheco-Barrios** et al. 2023. “Trends in Spinal Orthosis Utilization Among Patients Insured through Medicare Part B.” *Spine*, November. <https://doi.org/10.1097/BRS.0000000000004875>.
32. Bharadwaj, H. R., Awuah, W. A., Adebuseye, F. T., Tan, J. K., Ali, S. H., **Pacheco-Barrios, N.**, & Papadakis, M. (2024). Awake craniotomies in South America: Advancements, challenges, and future prospects. *Journal of Central Nervous System Disease*, 16, 11795735241238681.
33. Lazo-Porras, Maria, Hueiming Liu, Menglu Ouyang, Xuejun Yin, Alejandra Malavera, Tiana Bressan, Wilmer Cristobal Guzman-Vilca, **Pacheco N** et al. 2022. “Process Evaluation of Complex Interventions in Non-Communicable and Neglected Tropical Diseases in Low- and Middle-Income Countries: A Scoping Review.” *BMJ Open* 12 (9): e057597.
34. Carrillo-Larco RM, Altez-Fernandez C, **Pacheco-Barrios N**, Bambs C, Irazola V, Miranda JJ, Danaei G, Perel P. Cardiovascular Disease Prognostic Models in Latin America and the Caribbean A Systematic Review. *Global Heart* 2019;14(1):81-93. <https://doi.org/10.1016/j.gheart.2019.03.001>
35. Mendoza-Saldaña, Juan-Diego, José Enrique Viton-Rubio, Susana Brissvany Guzman-Carrasco, **Niels Victor Pacheco-Barrios**, and Carla Rebeca Lainez-Casal. 2021. “Characteristics and Outcomes from COVID-19 among Peruvian Physicians: A Nationwide Register-Based Study.” *Archives of Environmental & Occupational Health*, December 1–5.
36. Saal-Zapata, Giancarlo, Aaron Rodriguez-Calienes, Marco Malaga, Victor Velasquez-Rimachi, Cristian Moran, Diego Bustamante-Paytan, **Niels Pacheco-Barrios**, Kevin Pacheco-Barrios, Carlos Alva-Díaz, and Melanie Walker. 2022. “Microsurgical and Endovascular Treatment of Posterior Inferior Cerebellar Artery Aneurysms: A Systematic Review and Meta-Analysis.” *Journal of Neurosurgical Sciences*, April. <https://doi.org/10.23736/S0390-5616.22.05710-1>.
37. Mejia CR, Chacón JI, Garay Jaramillo E, Jorge Torrealba M, Delgado-García S, Aveiro Róbaló R, **Pacheco-Barrios N**, Serrano F. Training and research carried out by human resources in health, Latin America. *Educación Médica*. 2018. <https://doi.org/10.1016/j.edumed.2018.08.005>
38. Nieto-Gutierrez W, Tellez WA, **Pacheco-Barrios N**, Aguirre-Tipismana L, Zevallos-Morales A, Taype-Rondan A. Características de patentes de dispositivos para rehabilitación física registradas en los Estados Unidos. *Revista Habanera de Ciencias Médicas*. 2019

39. Christian Richard Mejia-Alvarez, Jhosselyn Chacon-I, Dayanne Benites-Gamboa, **Niels Pacheco-Barrios**, Giancarlo F. Castillo-Tarrillo, Dercy Centeno-Leguia, Joseph Wendell-Cubilla, J. Antonio Grandez-Urbina. Frequency and associated factors related to sexual addiction in Medical Students from 16 Latin American Cities, 2016: a regional multicentric study version. *F1000Research* 2020, 9:108
<https://doi.org/10.12688/f1000research.21418.1>
40. Rodrigo M Carrillo-Larco, C Joel Benites-Moya, Cecilia Anza-Ramirez, Leonardo Albitres-Flores, Diana Sánchez-Velazco, **Niels Pacheco-Barrios**, Antonio Bernabe-Ortiz **A systematic review of population-based studies on lipid profiles in Latin America and the Caribbean.** *eLife* 2020;9:e57980 DOI: 10.7554/eLife.57980
<https://elifesciences.org/articles/57980>
41. Alarcon-Braga, Esteban A., Farley E. Salazar-Valdivia, Jose M. Estrada-Grossmann, Carolina Mendez-Guerra, **Niels Pacheco-Barrios**, and Ali Al-Kassab-Córdova. 2023. "Pre-Extensively Drug-Resistant and Extensively Drug-Resistant Tuberculosis in Latin America and the Caribbean: A Systematic Review and Meta-Analysis." *American Journal of Infection Control*, December. <https://doi.org/10.1016/j.ajic.2023.12.001>.
42. Vidyasagaran AL, Ayesha R, Boehnke JR, Kirkham J, Rose L, Hurst JR, et al. Core outcome sets for trials of interventions to prevent and to treat multimorbidity in adults in low and middle-income countries: the COSMOS study. *BMJ Glob Health*. 2024 Aug 19;9(8):e015120.
43. Vidyasagaran, A. L., Ayesha, R., Boehnke, J., Kirkham, J., Rose, L., Hurst, J., Miranda, J. J., Rana, R. Z., Vedanthan, R., Faisal, M., Siddiqi, N., & COSMOS Collaboration. (2024). Core outcome sets for trials of interventions to prevent and to treat multimorbidity in low- and middle-income countries: the COSMOS study. *medRxiv : The Preprint Server for Health Sciences*. <https://doi.org/10.1101/2024.01.29.24301589>
44. Bharadwaj, H. R., A. A. Wireko, F. T. Adebuseye and **N Pacheco-Barrios** 2023. "Challenges and Opportunities in Prostate Cancer Surgery in South America: Insights into Robot-assisted Radical prostatectomies—A Perspective." *Health Science*. <https://onlinelibrary.wiley.com/doi/abs/10.1002/hsr2.1519>.
45. Kian Daneshi, Hamid Reza Khademi Mansour, **Niels Pacheco-Barrios**, Ayobami Asaju, Mauricio Perez Pachon, Alfredo Hoyos, Ankur Khajuria, A Bibliometric Analysis of the Top 100 Papers on Gluteal Augmentation, *Aesthetic Surgery Journal Open Forum*, 2024;, ojae053, <https://doi.org/10.1093/asjof/ojae053>
46. Martínez-Esteban, P. C., & Guillén-Pinto, D. (2022). Guía de práctica clínica para el diagnóstico y tratamiento de la distrofia muscular de Duchenne. Sociedad Peruana de Neurología. Versión Corta. *Acta Médica*. http://www.scielo.org.pe/scielo.php?script=sci_arttext&pid=S1728-59172022000200205
47. García-Solorzano, F. O., & **Pacheco-Barrios, N.** (2021). Aceptación de la vacunación contra la COVID-19 durante la primera ola pandémica en Perú. *Revista Del Cuerpo*. http://www.scielo.org.pe/scielo.php?pid=S2227-47312021000300022&script=sci_arttext
48. Calisaya-Madariaga IG, Serrano MNM, Cruzado-Saucedo JL, Común-Gutierrez DK, Abanto-Florez LM, Acurio K, **Pacheco-Barrios, N.** Challenges and opportunities in neurosurgery in Peru: disparities, research, and global collaboration. *Neurosurg Rev*. 2024 Sep 12;47(1):595.
49. Khajuria A, Mansour HRK, Muhammad I, Asare A, Tammasse I, Suresh J, **Pacheco-Barrios, N** et. al. Tranexamic Acid in Rhinoplasty and Septoplasty: A Systematic Review and Meta-analysis of Randomized Controlled Trials. 2024 Nov 1;12(11):e6275.
50. Barbagli G, Soto-Rubio D, Pacheco-Barrios N, Li C, Hussein A, Kelbert J, **Pacheco-Barrios, N**, et al. Harlequin sign due to an upper thoracic paravertebral lesion. A systematic review of the literature. 2024 Nov 1;129:110850.

51. Marin-Castañeda LA, Gómez-Villarroel RA, Pacheco Aispuro G, Palomera-Garfias N, **Pacheco-Barrios N**, Sandoval-Orellana VM, et al. Comparative efficacy and safety of external ventricular drains and intraparenchymal pressure monitors for intracranial pressure monitoring in traumatic brain injury: A systematic review and meta-analysis. *Neurocrit Care*. 2024 Oct 2;1–13.
52. Palavani LB, Mitre LP, Camerotte R, Nogueira BV, Canto GL, Chen H-C, **Pacheco-Barrios N**, et al. Advancements and challenges: immunotherapy therapy in high-grade glioma - a meta-analysis of randomized clinical trials. *J Neurooncol*. 2024 Sep 4;1–11.
53. Effect of occupational therapy on cognition in patients with mild Alzheimer’s Disease: A scoping review. **Niels Pacheco-Barrios**, Leandra Ramin-Wrighth, Sandra Zhong, Marion Stokvis-Blok, Alanna Barrios-Ruiz, Aala Elhadi, Stefany Alfaro-Amez, Deborah Estrella, João P. G. Kasakewitch, Cecilia Plaza, Camila Garcia, Renata Medeiros, Nayara Rutes, Guilherme Areas. *PPCR Journal*.
54. **DECREASE**: decrease cognitive decline in Alzheimer’s disease through occupational therapy a randomized controlled phase III trial protocol. **Niels Pacheco-Barrios**, Leandra Ramin-Wrighth, Sandra Zhong, Marion Stokvis-Blok, Alanna Barrios-Ruiz, Aala Elhadi, Stefany Alfaro-Amez, Deborah Estrella, João P. G. Kasakewitch, Cecilia Plaza, Camila Garcia, Renata Medeiros, Nayara Rutes, Guilherme Areas. *PPCR Journal*.
55. Estrada-Grossmann JM, Alarcon-Braga E, Salazar-Valdivia F, Silvana-Jauregui A, Acurio K, **Pacheco-Barrios N**, et al. Risk Factors for Malignant Progression to Adult-T-cell Leukemia/Lymphoma in HTLV-1 Carriers: A Systematic Review of Cohort and Case-Control Studies Risk Factors for ATL in HTLV 2025 Apr 19; Available from: <https://www.sciencedirect.com/science/article/pii/S2152265025001442>
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Conference papers

65. Pacheco-Barrios N, Riveros J, Quintana-García LA, Flores SM, Butrón-Verástegui VE, Itusaca N, et al. 1040 safety of the use of a handmade external ventricular drain versus a classic prefabricated one: A retrospective cohort. *Neurosurgery* [Internet]. 2025 Apr 1;71(Supplement_1):142. Available from: http://dx.doi.org/10.1227/neu.0000000000003360_1040
66. Canazas-Paredes, F., Pacheco, N., Acurio, K., Roman, F. V., Calisaya-Madariaga, I., Zambrano, D., Alarcon-Braga, E., Fregni, F., & Pacheco-Barrios, K. (2024). Quality of life (QoL) following traumatic brain injury (TBI) in Latin America: A systematic review (P7-11.014). *Neurology*. <https://doi.org/10.1212/wnl.0000000000206645>
67. Calisaya-Madariaga, I., Acurio, K., Pacheco, N., & Arista, M. C. (2024). Hereditary myopathy with early respiratory failure (HMERF): A case report and review of the literature (P3-11.014). *Neurology*. <https://doi.org/10.1212/wnl.0000000000206198>
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89. M325LBA Pacheco-barrios K, **Pacheco-Barrios N**, Carlos A-D, Victor V. Efficacy of the Use of Dihydroergotamine in the Management of Migraine Attacks: A Systematic Review with Meta-Analysis. 143rd Annual Meeting American Neurological Association. 2018 <https://doi.org/10.1002/ana.25331>
90. Worldwide Frequency of Phantom Limb Pain in People with Amputations: A Systematic Review and Meta-Analysis. 145th Annual Meeting American Neurological Association. Kevin Pacheco-Barrios, Damaris Ramirez-Alva, Ana Balbuena-Pareja, Niels Pacheco-Barrios, Stefano Giannoni-Luza, Anna Carolina Gianlorenço, Luna Vasconcelos Felipe, Ivan Chacha-Angulo, Felipe Fregni. 2020 <https://doi.org/10.1002/ana.25865>
91. Global Prevalence of Tuberculosis in the Central Nervous System: A Systematic Review and Meta-Analysis. 145th Annual Meeting American Neurological Association. Jose Ernesto Fernández-Chinguel, Niels Pacheco-Barrios, Martina Guillermo-Roman, Aaron Rodriguez Calienes, Kevin Pacheco-Barrios. 2020 <https://doi.org/10.1002/ana.25865>
92. Gluteal Fat Grafting: A Bibliometric Analysis of the Top 100 most cited articles. 92nd Plastic Surgery annual meeting. Niels Pacheco-Barrios, Yong Yie Liew, Nivan Lakshman, Alfredo Hoyos-Ariza, Ankur Khajuria. Accepted for oral presentation in October 2023
93. Factors Associated with capsular contracture following breast augmentation: an umbrella review of systematic reviews. 92nd Plastic Surgery annual meeting. Riccardo Conci, Niels Pacheco-Barrios, Yong Yie Liew, Nivan Lakshman, Qiming Dong, Ankur Khajuria. Accepted for oral presentation in October 2023.
94. Alva-Diaz, C., Julca, M. M., Martinez-Esteban, P., Ballon, P. M., Ecos, F. C., & Pacheco-Barrios, N. (2021). Molecular diagnosis of Duchenne Muscular Dystrophy: Systematic review and meta-analysis. *Journal of the Neurological Sciences*, 429(118404), 118404.
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100. Galarza MMC, Paredes NC, Vascones-Roman FF, Vera D, Perez SAA, Calisaya-Madariaga I, et al. Efficacy and Safety Outcomes of thrombolysis for stroke in Low-and Middle-Income countries: Systematic Review and Metanalysis. 2025 Feb 1;56. Available from: https://scholar.google.com/citations?view_op=view_citation&hl=en&citation_for_view=e4jKwLwAAAAJ:UHK10RUVsp4C
101. Pacheco-Barrios N, Riveros J, Quintana-García LA, Flores SM, Butrón-Verástegui VE, Itusaca N, et al. 1040 safety of the use of a handmade external ventricular drain versus a classic prefabricated one: A retrospective cohort. *Neurosurgery* [Internet]. 2025 Apr 1;71(Supplement_1):142–142. Available from: http://dx.doi.org/10.1227/neu.0000000000003360_1040
102. Howard C, Reich M, Lou L, **Pacheco-Barrios N**, Alterman R, Ríos A, et al. A.2 Cognitive outcomes of deep brain stimulation depend upon subiculum connectivity. *Canadian Journal of Neurological Sciences / Journal Canadien des Sciences Neurologiques* [Internet]. 2025 Jun 1;52(s1):S3–S3. Available from: <http://dx.doi.org/10.1017/cjn.2025.10148>

Clinical Guidelines or Health Technology assessments

103. (*Clinical Guideline*) Guía de práctica clínica para el diagnóstico y tratamiento de la distrofia muscular de duchenne y becker. Peruvian Society of Neurology, from [GPC - Distrofia M Duchenne- VERSIÓN USUARIO.pdf](http://www.gpc-distrofia-m- Duchenne- VERSION USUARIO.pdf) (spneurologia.org.pe).
104. (*Health Technology assessments*) Evaluación de Tecnología Sanitaria N° 023-DETS-IETSI-2023. Eficacia y seguridad de la prótesis de resección del fémur proximal recubierto en plata, para el tratamiento de pacientes con tumores ubicados en el fémur proximal. Retrieved July 29, 2024, from <https://repositorio.essalud.gob.pe/handle/20.500.12959/4261>

Podcast episodes – Stimulating Brains

105. Pacheco, Niels; Taha, Alaa; Horn, Andreas (2023). #41: Christelle Baunez – The role of the subthalamic nucleus in brain stimulation. figshare. Media. <https://doi.org/10.6084/m9.figshare.24247408>
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107. Pacheco, Niels; Horn, Andreas (2023). #45: Mac Shine and Paul Cisek - Exploring the evolution, integration and complexities of the brain: basal ganglia, dopamine, and beyond. figshare. Media. <https://doi.org/10.6084/m9.figshare.24762297>
108. Pacheco, Niels; Horn, Andreas (2024). #49: Dora Hermes & Kai Miller – Discovering the secrets of the brain by invasive stimulation & signal recording. figshare. Media. <https://doi.org/10.6084/m9.figshare.25422001>
109. Pacheco, Niels; Horn, Andreas (2024). #55: Jens Volkmann – Image Guidance, Disease Modification and why there may be no ‘sweet spot’ for DBS.. figshare. Media. <https://doi.org/10.6084/m9.figshare.26166421>
110. Pacheco, Niels; Horn, Andreas (2024). #57: Mallory Hacker & David Charles – Could DBS be slowing motor progression if applied very early?. figshare. Media. <https://doi.org/10.6084/m9.figshare.26381401>
111. Pacheco, Niels; Horn, Andreas (2024). #60: Dr. Nanthia Suthana – Pushing Boundaries: Memory Enhancement, Virtual Reality, and Trauma Therapy in Neuroscience. figshare. Media. <https://doi.org/10.6084/m9.figshare.27321225.v1>
112. Pacheco, Niels; Horn, Andreas (2024). #65: Jim McNasby - Transforming Lives with DBS and Advocacy in Parkinson's Disease. figshare. Media. <https://doi.org/10.6084/m9.figshare.28075016.v3>

113. Pacheco, Niels; Horn, Andreas (2025). #69: Todd Harrington – Adaptive DBS and the Future of Neuromodulation: Insights from the ADAPT PD Trial and Beyond. figshare. Media. <https://doi.org/10.6084/m9.figshare.28578743.v1>
114. Pacheco, Niels; Horn, Andreas (2025). #70: Rees Cosgrove – When Neurosurgery Chooses You: Rediscovering the Lost Art of Lesions and the Future of Brain Stimulation. figshare. Media. <https://doi.org/10.6084/m9.figshare.28840943.v1>
115. Pacheco, Niels; Horn, Andreas (2025). #72: John Rolston — Closed-Loop Stimulation, Seizure Dynamics, and the New Frontiers of Epilepsy Treatment. figshare. Media. <https://doi.org/10.6084/m9.figshare.29286704.v1>

VOLUNTER EXPERIENCE

- Co-founder** *Synopsis MHN (non-profit organization) – Lima Perú* 01/2018- Current
- Reduction of mental health stigma through education and promotion of good mental health behaviors in general population.
 - Develop research and interventions related to neuropsychiatric fields.
 - Design interventions to improve medical education related to neuropsychiatric fields.
- Founder** *Astrocyte, feeding your learning (non-profit organization) – Lima Perú* 11/2020- Current.
- Create interventions to improve medical education in Peru and across the world.
 - Develop research and interventions related to medical education.
 - Design programs of clinical, research and career mentorship to enhance the development of future physicians.
- Director** *Neurosurgery Journal Club, Journal Club UPOCH – Lima Perú* 08/2022- 07/2023
- Discuss ideas and articles in Neurosurgery.
 - Educate participants on how to read a research article.
 - Educate participants in Neurosurgical principles.
- Director** *Medical Education Journal Club, Journal Club UPOCH – Lima Perú* 01/2022- 07/2022.
- Discuss ideas and articles in Medical Education.
 - Educate participants on how to read a research article.
 - Educate participants in Medical Education principles.
- Volunteer** *Mission Brain Cayetano– Lima Perú* 02/2022- 12/2022.
- Mission Brain Chapter at Cayetano Heredia Peruvian University.
 - Nonprofit organization with the goal of providing neurosurgical expertise and resources to patients, caregivers, and healthcare providers in underserved areas throughout the world.
- Fellow** *REDECS, network of Clinical efficiency – Lima Perú* 04/2020- 07/2023
- Development of Clinical practice guidelines using synthesis of medical evidence. This includes systematic review methods and GRADE methodology to obtain certainty of the evidence and finally give recommendations in the diagnosis and treatment of a diseases in the particular clinical context of the required institution.
- Volunteer** *Gender equality initiative in Global Surgery (GEIGS)* 08/2020 – 06/2021
- Organization related to Harvard Paul Farmer Global Surgery Fellowship that seeks to empower women in global surgery.
 - Participant in journal club for scientific articles, conferences, writing short articles and mentoring programs.
- Learning Mananger, Young Peruvian Leaders Lima (YPL)** 03/2020- 12/2020
- Non-profit organization – Lima, Peru*
- Empower young adults to adopt a proactive mindset about their local problems.
 - Developed strategies and organized activities to achieve the Sustainable Development Objectives made by United Nations.

Volunteer, Nutre-Perú

03/2019- 12/2019

Cayetano Heredia Peruvian University – Lima, Peru

- Empower women and fight malnutrition and anemia among children 0-5 years in rural Peruvian communities.
- Developed sustainable strategies and organized activities to conduct in a twice-a-year trip to a rural community, holding weekly meetings with a multidisciplinary team.

Volunteer, Medical Missions to Peru

01/2015, 01/2016, 07/2018

- 3-day medical mission to screen and provide basic medical care in a rural community.
- Medical Missions to Peru is a common effort of the CLM in the US and Peru with Solidarity Experiences Abroad (SEA) from Canada that began in 2004.

Director Assistant, SCORE-UPCH, IFMSA

03/2017-12/2017

Cayetano Heredia Peruvian University – Lima, Peru

- Research exchange committee (SCORE) of the international federation of medical student's association (IFMSA).
- Develop research collaboration networks and exchange for students at UPCH and around the world.
- Train medical students in basic knowledge related to research.

Director Assistant, SCOPH-UPCH, IFMSA

03/2016-07/2016

Cayetano Heredia Peruvian University – Lima, Peru

- Public Health committee (SCOPH) of the international federation of medical student's association (IFMSA).
- Promotion of a healthy lifestyle among medical students and the general community.

Subdirector and Mentor, SOCEMCH (Sociedad Científica de estudiantes de Medicina)

03/2015- 12/2021

Cayetano Heredia Peruvian University – Lima, Peru

- Train medical students in epidemiology and biostatistics to develop medical research.
- Develop student collaboration groups to conduct research.

Volunteer, SECC (Sociedad de Ciencias Clínicas)

03/2018- 08/2022

Cayetano Heredia Peruvian University – Lima, Peru

- Teach medical knowledge to medical students of lower years. I participated at the Neurophysiology and Nephrology course.

COURSES AND CONFERENCES ATTENDED

- **Conferences**

- I International Congress on Pneumology, Lima-Peru; 07/2015
- 143rd Annual Meeting of the American Neurology Association, Atlanta 2018; 10/2018
- 144th Annual Meeting of the American Neurology Association, Los Angeles 2019; 10/2019
- 145th Annual Meeting of the American Neurology Association; 10/2020
- AAN 2023. Annual Meeting – Winner a Travel Grant
- AES 2023 American Epilepsy Society – Oral presentation
- AANS 2023. Annual Meeting
- ASSFN Bienal meeting 2023 – Winner of a Travel Grant

- **Given Lectures**

- Junior Research toolbox – SOCEMCH Research principles elective April 2020, 2021, 2022.

- The SOCIMEP experience – SOCIMEP, June 24, 2020
- Creating a Research profile – ADIECS San Marco Major University (UNMSM), May 29, 2021
- Sharing my research to increase citations – SOCEMCH Research principles elective December 2020
- Journal Club in Neurosurgery – SIRAY Surgery and Global Health, October 2020
- Cross over Neurosurgery and Artificial Intelligence – Journal Club UPCH, February 8, 2023
- **Courses**
 - Systematic reviews and meta-analysis. High yield UK. December 2022.
 - Concept to publication. High yield UK. July 2022.
 - Systematic reviews and meta-analysis. The Match Guy. January 2023.
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